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SEZG628T : DISSERTATION OUTLINE

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DISSERTATION / PROJECT / PROJECT WORK TITLE	AI-Driven Relevance-Based Retrieval Augmented Generation Framework for Application Development Context

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Dissertation / Project / Project Work Title: AI-Driven Relevance-Based Retrieval Augmented Generation Framework for Application Development Context

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Dissertation / Project /Project Work Done by:

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Degree Program: M.Tech

Research Area: Artificial Intelligence

Dissertation / Project Work carried out at:

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1. Broad Area of Work

Artificial Intelligence has become a core driving force in research and industry, particularly in the domains of information retrieval, natural language processing, and software engineering. The foundation of this project lies at intersection of these disciplines, focusing on how intelligent systems can reduce cognitive load and improve developer productivity by combining fragmented knowledge sources.

The broad academic areas relevant to this research include:

- Natural language processing: To enable the system to understand developer queries expressed in natural language and map them to relevant artifacts.
- Knowledge representation: To structure application development artifacts in vector databases and metadata frameworks to support efficient retrieval and contextual reasoning.
- Automated reasoning: To combine retrieved knowledge with stored information to resolve dependencies, identify conflicts and generated actionable insights for developers.
- Information Retrieval: Developing semantic search and relevance-based retrieval techniques to locate the most contextually appropriate artifacts from requirements, tickets and video recordings.
- Software Engineering: Applying AI techniques to real-world development workflows, including requirement analysis, design, coding, and testing to enhance continuity and reduce bottlenecks.

2. Background

Software development projects generate a wide variety of artifacts including requirement documents, JIRA tickets, bugs, video recordings. These resources are often scattered across different tools and repositories, which is time consuming for developers to quickly locate the information they need. This fragmentation increases cognitive load, as developers must spend significant effort piecing context together. The challenge becomes more pronounced when knowledge transfer is incomplete or dependent on the availability of specific individuals. Even when recordings or documentation exist, doubts often arise later, and the person best positioned to clarify them may no longer be available. For newly joined developers, the situation is even more demanding. They must

shift from knowledge transfer recordings, documentation and artifacts to understand the product, but when questions arise, they often depend on colleagues for clarifications or spend additional time searching through documents. Moreover, not all developers are equally familiar with every component. Dependencies across modules frequently require clarifications that again depend on the availability of others, creating bottlenecks in continuity and productivity.

Traditional keyword-based search methods are inadequate in this setting because they fail to capture the semantic meaning of queries or connect related artifacts across sources. Advances in Artificial Intelligence, particularly in Natural Language Processing and Large Language Models, offer new possibilities by enabling systems to interpret natural language queries and generate coherent responses.

However, these models alone are limited by their training data and cannot reliably provide context-aware answers grounded in project artifacts. This gap has led to growing interest in Retrieval-Augmented Generation framework, which combines semantic retrieval with generative models. By embedding the artifacts into a vector database enriched with metadata, and retrieving them based on relevance, such a system can act as an intelligent assistant for developers, thereby reducing cognitive load, improving continuity and ensuring that knowledge remains accessible and actionable even in dynamic environments where teams evolve and modules change.

3. Objectives

The objectives of my project are as follows:

- To analyze the current state knowledge access in application development and the extent of individual reliance of scattered project resources for clarifications.
- To design an ingestion pipeline and retrieval pipeline that combines the heterogenous project artifacts into an integrated knowledge base.

- To develop a relevance-based Retrieval-Augmented Generation framework that delivers coherent, context-aware answers to developer queries.
- To evaluate and validate the framework using metrics such as relevance of retrieval, consistency of outputs and reduction in query resolution time compared to manual search methods.

4. Scope of Work

The scope of this dissertation is to develop a relevance-based Retrieval-Augmented Generation framework that consolidates heterogenous project artifacts such as requirement documents, JIRA tickets, knowledge transfer video recordings into a unified knowledge base. The framework will implement an ingestion pipeline for continuous integration of artifacts, a semantic retrieval mechanism for effective context relevance and an orchestration layer to ensure coherent, relevance, consistency outputs and reduction in query resolution time compared to manual search, thereby demonstrating its efficiency in supporting developer continuity and reducing cognitive load.

5. Plan of Work

Phases	Start Date-End Date	Work to be done
Dissertation Outline	17 Jan 2026 – 7 Feb 2026	Prepare Abstract report, review with Supervisor and Additional Examiner and outline report submission
Design & Development	8 Feb 2026 – 8 Mar 2026	Completion the design and development of the part of the proposed framework
Mid Semester Report	9 Mar 2026 – 28 Mar 2026	Mid-semester report review with Supervisor and Additional Examiner and report submission
Testing	29 Mar 2026 – 17 Apr 2026	Execution of testing activities and validation of framework components
Dissertation Submission	18 Apr 2026 – 12 May 226	Final Review and submission of Dissertation

6. Literature References

The literature referred to in this dissertation includes both foundational theories and recent advances in Retrieval-Augmented Generation, developer workflows, and supporting technical modules.

[1] N. Belkin, "Anomalous states of knowledge as a basis for information retrieval," Canadian Journal of Information Science, vol. 5, pp. 133–143, 1980.

[2] S. Avula, G. Chadwick, J. Arguello, and R. Capra, "Searchbots: User engagement with chatbots during collaborative search," in Proc. 2018 Conf. Human Information Interaction & Retrieval, pp. 52–61, ACM, 2018.

[3] M. Baqar, RAG4Tickets: AI-Powered Ticket Resolution via Retrieval-Augmented Generation on JIRA and GitHub Data, Cisco Systems Inc., 2018.

[4] K. Nguyen, Context-Based User Story Enhancement with LLM and RAG: A Multi-Agent System Implementation, Master's Thesis, Tampere University, 2025.

[5] A. S. V. Vardhan, S. S. Chamarthy, V. S. Balasubramaniam, M. S. R. Prasad, S. Kumar, and S. Sangeet, Exploring RAG and GenAI Models for Knowledge Base Management, Southern New Hampshire University & Columbia University, 2024.

[6] J. J. Pan, J. Wang, and G. Li, "Survey of Vector Database Management Systems," arXiv preprint arXiv:2310.14021, 2023.

[7] J. Richman, "Data ingestion pipeline: Types, key concepts, & use cases," Estuary Technical Report, 2024.

[8] H. Tahir, "9 Best LLM Orchestration Frameworks for Agents and RAG," ZenML Blog, Oct. 2025

7. Particulars of the Supervisor and Examiner

	Supervisor	**Additional Examiner**
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8. Remarks of the Supervisor

The study and project seem to very promising. The student had covered all the relevant literature review. The project is holding the current domain of research and justified the level of dissertation for master studies. The outcome covered by the project seems very interesting and include all the state of arts in today's era of information science. The project also includes the user study on the working prototype. I approve the following project as the supervisor.

Information about the Supervisor: The dissertation has been carried out under the supervision of Mr.Vaideeshwaran, who holds a B.Tech degree in Information Technology and has over 17 years of professional experience in software industry. His broad expertise and insights into applied computing have provided valuable direction in shaping the scope and conduct of this research.